Practical Exam: Object-Oriented Programming in Java

Exam Guidelines

- Time Limit: 60 minutes
- Language: Java
- Instructions: Create a folder “YourLastname+PracticalExamB”. Implement the required classes according to the specifications below. Save your file using the main class name provided in each set.

Consultant Hierarchy & Fee Payment

Objectives

Demonstrate understanding of inheritance, constructor chaining using `super`, and implementing an interface.

Specifications

1. **Create an interface** named `Payable`:
    - Method signature: `double computePay();`
2. **Create a superclass** named `Personnel`:
    - Field: `protected String name`
    - Parameterized constructor that initializes `name`.
3. **Create a subclass** named `Consultant` that extends `Personnel` and implements `Payable`:
    - Field: `private double professionalFee`
    - Parameterized constructor `Consultant(String name, double professionalFee)` that invokes `super(name)` and sets `professionalFee`.
    - Override `computePay()` to return `professionalFee`.
    - A helper method `public void displayConsultant()` that prints the consultant's name and fee.
4. **Create the main class** named `PracticalExamB` with a `main` method:
    - Instantiate a `Consultant` object with the name `"Eli"` and a fee of `22000.00`.
    - Call `displayConsultant()` on the object.

Expected Output:

```
Consultant: Eli
Fee: 22000.00
```

Grading Rubric (Total: 20 Points)

Criteria	Points	Description
Class & Inheritance Setup	6 pts	Correct usage of class, extends, abstract, or implements.
Constructors & super	5 pts	Proper constructor definitions and correct use of super(...) as the first statement.
Method Implementation	5 pts	Correct method signature, return types, and overriding.
Compilation & Output	4 pts	Program compiles without errors and reproduces the exact expected output.